Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

High-Performance Computing Lab for CSE **2026**

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Solution for Project 7

Due date: **Friday** 29 May 2026, 18:00 (Europe/Zurich)

HPC Lab for CSE 2025 — Submission Instructions

Following these instructions is mandatory: Submissions that do not comply **will not be considered**.

- **Submission Platform**: Assignments must be submitted via SAM-UP.
- **Required Files**: Your submission must include:
  - **Report**: A PDF summarizing your results and observations, based on this LaTeX template.
    - File name: `project_number_lastname_firstname.pdf`
    - You are allowed to discuss all questions with anyone you like; however: (i) your submission must list anyone you discussed problems with and (ii) you must write up your submission independently.
  - **Source Code Archive**: A **compressed tar archive**, named `project_number_lastname_firstname.tar.gz`, containing:
    - All source code files.
    - Build files and scripts (e.g., `Makefile`, batch job scripts, plot scripts, etc). If you modified any provided files, ensure they still work as expected.
    - The full LaTeX source of your report including all needed files to build the report (figures, listings, etc).
  - **Size Limit**: The total submission (PDF + archive) must not exceed **25 MB**.
  - **File Naming Rules**: Use only `ASCII` characters (avoid spaces, special symbols, and non-English letters).
- **Grading Process**: The TAs will evaluate your project based on:
  - Your report write-up.
  - Your code implementation.
  - Benchmarking your code's performance.

1. GPU architecture and performance characteristics [15 Points]

1.1. Peak performance [5 Points]

Using the provided utility `deviceQuery` we can retrieve the characteristics of the NVIDIA GeForce RTX 2080 Ti GPU. The output of the utility is reported in listing 1.

Listing 1: Output of the `deviceQuery` CUDA sample.

```
 CUDA Device Query (Runtime API) version (CUDART static linking)

Detected 1 CUDA Capable device(s)

Device 0: "NVIDIA␣GeForce␣RTX␣2080␣Ti"
  CUDA Driver Version / Runtime Version          13.0 / 13.0
  CUDA Capability Major/Minor version number:    7.5
  Total amount of global memory:                 10821 MBytes (11346575360
      bytes)
  (068) Multiprocessors, (064) CUDA Cores/MP:    4352 CUDA Cores
  GPU Max Clock rate:                            1545 MHz (1.54 GHz)
  Memory Clock rate:                             7000 Mhz
  Memory Bus Width:                              352-bit
  L2 Cache Size:                                 5767168 bytes
  Maximum Texture Dimension Size (x,y,z)         1D=(131072), 2D=(131072,
     65536), 3D=(16384, 16384, 16384)
  Maximum Layered 1D Texture Size, (num) layers  1D=(32768), 2048 layers
  Maximum Layered 2D Texture Size, (num) layers  2D=(32768, 32768), 2048
     layers
  Total amount of constant memory:               65536 bytes
  Total amount of shared memory per block:       49152 bytes
  Total shared memory per multiprocessor:        65536 bytes
  Total number of registers available per block: 65536
  Warp size:                                     32
  Maximum number of threads per multiprocessor:  1024
  Maximum number of threads per block:           1024
  Max dimension size of a thread block (x,y,z): (1024, 1024, 64)
  Max dimension size of a grid size    (x,y,z): (2147483647, 65535, 65535)
  Maximum memory pitch:                          2147483647 bytes
  Texture alignment:                             512 bytes
```

From listing 1 we can derive that:

- The number of Streaming Multiprocessors (SMs) is $n_{\text{SMs}} = 68$
- The number of CUDA cores per SM is $n_{\text{cores per SM}} = 64$
- The GPU maximumclock frequency is $f = 1.545$ GHz
- The CUDA Compute Capability is 7.5

From [**hpc˙cuda˙best˙practices˙guide**] we can derive the following information about the number of floating point operations per FMA.

- For single precision (FP32) FMAs $TP_{\text{FP32}} = \frac{64}{64} = 1$
- For double precision (FP64) FMAs $TP_{\text{FP64}} = \frac{64}{2} = 32$

Thus we can compute the theoretical peak performance of an NVIDIA GeForce RTX 2080 Ti for single and double precision as:

$$
\begin{aligned}
P_{\text{FP32}} &= n_{\text{SMs}} \cdot \frac{n_{\text{cores per SM}}}{TP_{\text{FP32}}} \cdot n_{FMA} \cdot f \\
&= 68 \cdot \frac{64}{1} \cdot 2\text{FP32} \cdot 1.545\text{GHz} = 13450,56 \text{ GFLOPS} \\
P_{\text{FP64}} &= n_{\text{SMs}} \cdot \frac{n_{\text{cores per SM}}}{TP_{\text{FP64}}} \cdot n_{FMA} \cdot f \\
&= 68 \cdot \frac{64}{32} \cdot 2\text{FP64} \cdot 1.545\text{GHz} = 420,33 \text{ GFLOPS}
\end{aligned}
$$

From Listing 1 we can get the following informations

- Global Memory : 10821 MiB = 10.57 GiB
- L2 Cache Size: 5.5 MiB
- L1 and Shared Memory per SM 96 KiB
- Read only memory 64 KiB
- Registers 64 Ki registers per block, each holding 4 bytes (total 256 KiB).
- Memory bus width: $W_{\text{mem}} = 352$ bits
- Memory clock frequency: $f_{\text{mem}} = 7$ GHz

So we can compute the (theoretical) peak memory bandwidth of a GPU as:

$$\begin{aligned} B_{\text{GPU}} &= \frac{W_{\text{mem}}}{8} \cdot f_{\text{mem}} \cdot 2 \\ &= \frac{352}{8} \cdot 7\text{GHz} \cdot 2 = 616 \text{ GB/s} \end{aligned}$$

1.2. Performance model: A simple roofline model [5 Points]

Compute ridge points

$$I_{\text{ridge}} = \frac{P_{\text{peak}}}{B_{\text{GPU}}}$$

For single precision:

$$I_{\text{ridge, FP32}} = \frac{13450.56 \text{ GFLOPS}}{616 \text{ GB/s}} = 21.83 \text{ GLOP/byte}$$

For double precision:

$$I_{\text{ridge, FP64}} = \frac{420.33 \text{ GFLOPS}}{616 \text{ GB/s}} = 0.68 \text{ GLOP/byte}$$

We can retrieve a roofline plot for the NVIDIA GeForce RTX 2080 Ti remembering that $P_{\text{attainable}} = \min P_{\text{peak}}, B_{\text{GPU}} \cdot I_{\text{ridge}}$.
In figure 1 we can see the roofline plot for the NVIDIA GeForce RTX 2080 Ti for single and double precision.

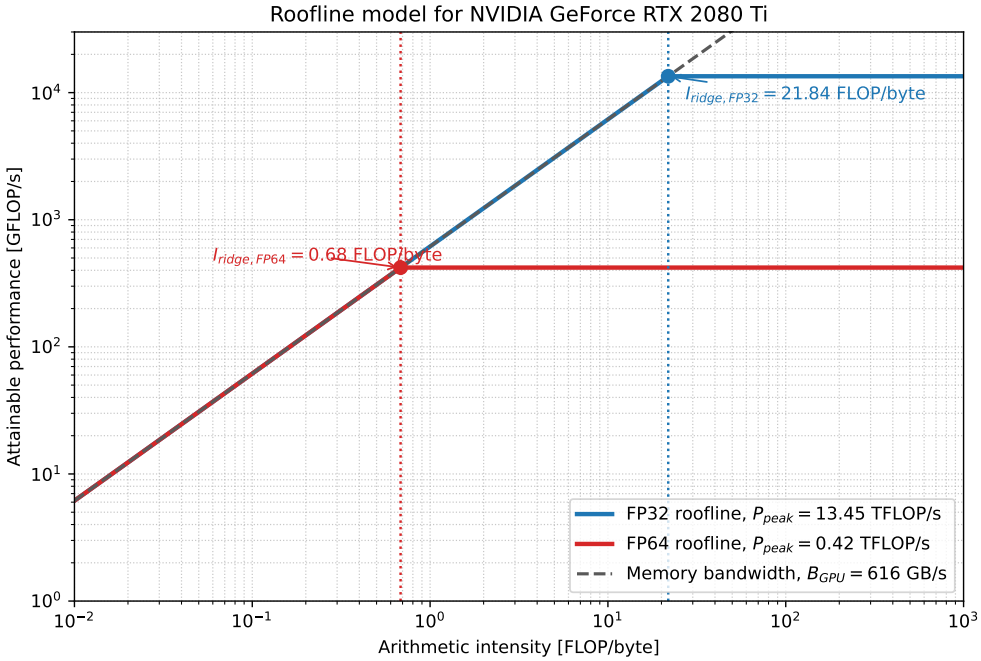


Figure 1: Roofline plot for the NVIDIA GeForce RTX 2080 Ti for single and double precision.

Thus, when the operational intensity of a kernel is lower than I_{ridge} the kernel is memory bound, otherwise it is compute bound.

2. Introduction to CUDA [50 Points]

2.1. Implement the simple kernels [30 Points]

The CUDA implementation of the functions in the file `simple_kernels.cu` is straightforward and it is based on the standard CUDA programming model. For everty function we first computed the correct index as `int i = blockIdx.x * blockDim.x + threadIdx.x` and then we check if the index is within the bounds of the input vectors by `if ( i < N )`. If the index is valid we perform the corresponding operation on the input vectors and we store the result in the output vector.
In this case, since in the `benchmark.cu` file the number of blocks is computed as `int blocks = (SIZE + threads - 1) / threads` and the number of threads per block is set to 256, we do not need in principle to worrry about the number of threads and blocks, since the kernel will be launched with enough threads to cover all the elements of the input vectors. However, we need to check if the index is within the bounds of the input vectors to avoid out-of-bounds memory accesses.

The output of the benchmark for the double format is reported in listing 2 and for the float format in listing 3.
From 2 we can retrieve the points in table 1 and from 3 we can retrieve the points in table 2.

Listing 2: Output of the simple kernels for the double format.

```
ripolo@eu-login-11:~/hpc_projects/project07/02cuda/simple_kernels$ srun --gpus=rtx_2080:1 ./benchmark
srun: job 171788 queued and waiting for resources
srun: job 171788 has been allocated resources
[benchmark] axpy: warmup=5 runs=25 avg_time=0.363086 ms
Compute utilization = 46.2073 GFLOPS/s
Memory utilization = 554.487 GB/s
first = 4, last = 4, expected = 4
[benchmark] add scaled diff: warmup=5 runs=25 avg_time=0.477494 ms
Compute utilization = 52.704 GFLOPS/s
Memory utilization = 562.176 GB/s
first = 1.6, last = 1.6, expected = 1.6
[benchmark] scaled diff: warmup=5 runs=25 avg_time=0.363736 ms
Compute utilization = 46.1247 GFLOPS/s
Memory utilization = 553.496 GB/s
first = 0.6, last = 0.6, expected = 0.6
[benchmark] scale: warmup=5 runs=25 avg_time=0.247412 ms
Compute utilization = 33.9054 GFLOPS/s
Memory utilization = 542.486 GB/s
first = 0.1, last = 0.1, expected = 0.1
[benchmark] linear combination: warmup=5 runs=25 avg_time=0.363924 ms
Compute utilization = 69.1512 GFLOPS/s
Memory utilization = 553.21 GB/s
first = 1.5, last = 1.5, expected = 1.5
```

Kernel	Compute Utilization (GFLOPS/s)	Arithmetic Intensity (FLOP/byte)
axpy	46.2073	$2/(3 \cdot 8)$
add scaled diff	52.704	$3/(4 \cdot 8)$
scaled diff	46.1247	$2/(3 \cdot 8)$
scale	33.9054	$1/(2 \cdot 8)$
linear combination	69.1512	$3/(3 \cdot 8)$

Table 1: Compute utilization and arithmetic intensity for the simple kernels in double precision.

Listing 3: Output of the simple kernels for the float format.

```
ripolo@eu-login-11:~/hpc_projects/project07/02cuda/simple_kernels$ srun --gpus=rtx_2080:1 ./benchmark
srun: job 172188 queued and waiting for resources
srun: job 172188 has been allocated resources
```

Kernel	Compute Utilization (GFLOPS/s)	Arithmetic Intensity (FLOP/byte)
axpy	91.7736	$2/(3 \cdot 4)$
add scaled diff	104.988	$3/(4 \cdot 4)$
scaled diff	91.798	$2/(3 \cdot 4)$
scale	67.0205	$1/(2 \cdot 4)$
linear combination	137.747	$3/(3 \cdot 4)$

Table 2: Compute utilization and arithmetic intensity for the simple kernels in single precision.

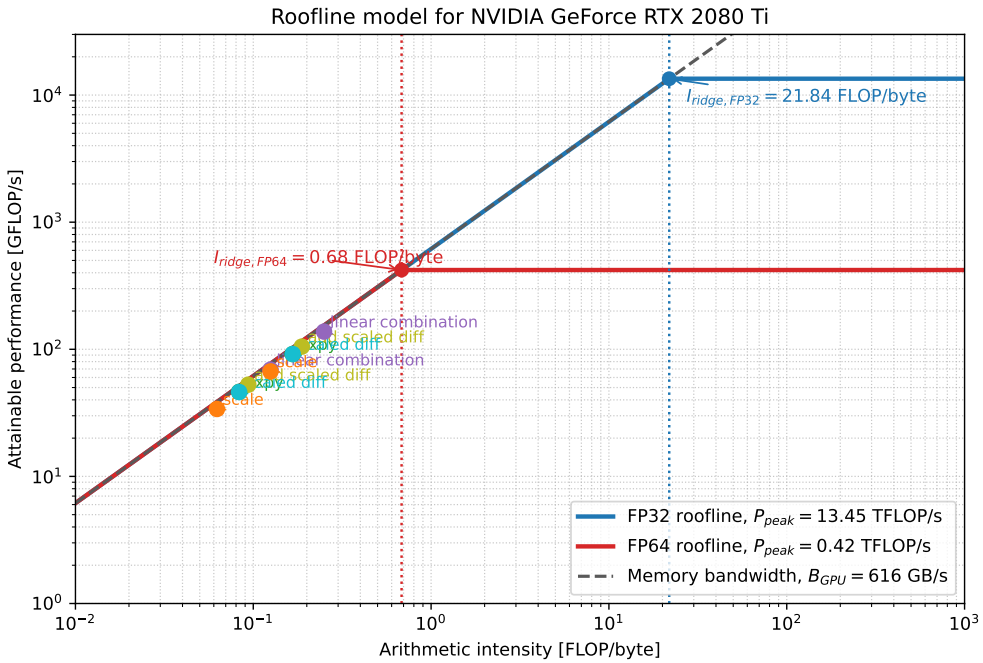


Figure 2: Roofline plot for the simple kernels in single and double precision. The points corresponding to the kernels are highlighted in red.

```
[benchmark] axpy: warmup=5 runs=25 avg_time=0.182811 ms
Compute utilization = 91.7736 GFLOPS/s
Memory utilization = 550.642 GB/s
first = 4, last = 4, expected = 4
[benchmark] add scaled diff: warmup=5 runs=25 avg_time=0.239703 ms
Compute utilization = 104.988 GFLOPS/s
Memory utilization = 559.933 GB/s
first = 1.6, last = 1.6, expected = 1.6
[benchmark] scaled diff: warmup=5 runs=25 avg_time=0.182762 ms
Compute utilization = 91.798 GFLOPS/s
Memory utilization = 550.788 GB/s
first = 0.6, last = 0.6, expected = 0.6
[benchmark] scale: warmup=5 runs=25 avg_time=0.125165 ms
Compute utilization = 67.0205 GFLOPS/s
Memory utilization = 536.164 GB/s
first = 0.1, last = 0.1, expected = 0.1
[benchmark] linear combination: warmup=5 runs=25 avg_time=0.182696 ms
Compute utilization = 137.747 GFLOPS/s
Memory utilization = 550.989 GB/s
first = 1.5, last = 1.5, expected = 1.5
```

In figure 1 we can see the roofline plot for the simple kernels in single and double precision. We can see that all the kernels are memory bound, since their operational intensity is lower than the ridge point of the roofline plot. However, we can see that the compute utilization of the kernels in single precision is higher than the compute utilization of the kernels in double precision, this is because the peak performance of the GPU for single precision is higher than the peak performance for double precision.

2.2. Implement the scalar product kernel [20 Points]

Listing 4: Output of the scalar product kernel for the double format.

```
ripolo@eu-login-20:~/hpc_projects/project07/02cuda/02_dot_product$ srun --gpus=rtx_2080:1 ./dot_prod
srun: job 191159 queued and waiting for resources
srun: job 191159 has been allocated resources
[benchmark] hpc_dot: warmup=5 runs=25 avg_time=27.3736 ms
Compute utilization = 1.22579 GFLOPS/s
Memory utilization = 9.80636 GB/s
Got: 16777216, expected: 16777216
```

3. Non-linear PDE mini-app with CUDA: The Fisher equation [35 Points]

output is

Listing 5: Output of the Fisher equation mini-app.

```
  ripolo@eu-login-20:~/hpc_projects/project07/03_fisher$ srun --gpus=rtx_2080:1 ./main 128 100 0.005
srun: job 192180 queued and waiting for resources
srun: job 192180 has been allocated resources
================================================================================

                       Welcome to mini-stencil!
version   :: CUDA
mesh      :: 128 * 128 dx = 0.00787402
time      :: 100 time steps from 0 .. 0.005
iteration :: CG 300, Newton 50, tolerance 1e-06
================================================================================

--------------------------------------------------------------------------------

simulation took 0.221755 seconds
1516 conjugate gradient iterations, at rate of 6836.37 iters/second
300 Newton iterations
--------------------------------------------------------------------------------

### 128, 100, 1516, 300, 0.221755 ###
Goodbye!
```

Using float the output cd